32 Non GATE CSE: Optimization (18)

32.1 Assignment Problem (2)

32.1.1 Assignment Problem: UGC NET CSE | June 2014 | Part 3 | Question: 59

The given maximization assignment problem can be converted into a minimization problem by

A. Subtracting each entry in a column from the maximum value in that column.
B. Subtracting each entry in the table from the maximum value in that table.
C. Adding each entry in a column from the maximum value in that column.
D. Adding maximum value of the table to each entry in the table.

ugcnetjune2014iii optimization assignment-problem

Answer key ☟

32.1.2 Assignment Problem: UGC NET CSE | Junet 2015 | Part 3 | Question: 67

In the Hungarian method for solving assignment problem, an optimal assignment requires that the maximum number of lines that can be drawn through squares with zero opportunity cost be equal to the number of

A. rows or columns
B. rows + columns
C. rows + columns -1
D. rows + columns +1

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Answer key ☟

32.2 Dual Linear Programming (1)

32.2.1 Dual Linear Programming: UGC NET CSE | December 2012 | Part 3 | Question: 24

If dual has an unbounded solution, then its corresponding primal has

A. no feasible solution
B. unbounded solution
C. feasible solution
D. none of these

ugcnetcse-dec2012-paper3 optimization dual-linear-programming

Answer key ☟

32.3 Linear Programming (1)

32.3.1 Linear Programming: UGC NET CSE | October 2020 | Part 2 | Question: 4

Consider the following linear programming (LP):

$$\begin{array}{ll} \text{Max.} & z = 2x_1 + 3x_2 \\ \text{Such that} & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 5 \\ & x_1, x_2 \geq 0 \end{array}$$

The optimum value of the LP is

A. 23
B. 9.5
C. 13
D. 8

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Answer key ☟

32.4 Linear Programming Problem (1)

32.4.1 Linear Programming Problem: UGC NET CSE | December 2013 | Part 3 | Question: 1

If the primal Linear Programming pro blem has unbounded solution, then it's dual problem will have

A. feasible solution
B. alternative solution
C. no feasible solution at all
D. no alternative solution at all

ugcnetcse-dec2013-paper3 optimization linear-programming-problem

Answer key

32.5 Optimization (8)

32.5.1 Optimization: UGC NET CSE | December 2012 | Part 3 | Question: 18

In a Linear Programming Problem, suppose there are three basic variables and 2 non-basic variables, then the possible number of basic solutions are

A. 6
B. 8
C. 10
D. 12

ugcnetcse-dec2012-paper3 optimization linear-programming

Answer key

32.5.2 Optimization: UGC NET CSE | December 2013 | Part 3 | Question: 2

Given the problem to maximize $f(x), X = (x_1, x_2, \ldots, x_n)$ subject to m number of in equality constraints. $g_i(x) \leq b_i$, i=1, 2, ... m including the non-negativity constrains $x \geq 0$. Which of the following conditions is a Kuhn-Tucker necessary condition for a local maxima at $\bar{x}$?

A. $\frac{\partial L(\bar{X}, \bar{\lambda}, \bar{S})}{\partial x_j} = 0, j = 1, 2 \ldots m$
B. $\bar{\lambda}_i [g_i(\bar{X}) - b_i] = 0, i = 1, 2 \ldots m$
C. $g_i(\bar{X}) \leq b_i, i = 1, 2 \ldots m$
D. All of these

ugcnetcse-dec2013-paper3 optimization linear-programming

Answer key

32.5.3 Optimization: UGC NET CSE | December 2013 | Part 3 | Question: 3

The following Linear Programming problem has:

Max $\quad Z = x_1 + x_2$

Subject to $\quad x_1 - x_2 \geq 0$

$\qquad 3x_1 - x_2 \leq -3$

and $\quad x_1, x_2 \geq 0$

A. Feasible solution
B. No feasible solution
C. Unbounded solution
D. Single point as solution

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Answer key

32.5.4 Optimization: UGC NET CSE | December 2015 | Part 3 | Question: 47

In constraint satisfaction problem, constraints can be stated as

A. Arithmetic equations and inequalities that bind the values of variables
B. Arithmetic equations and inequalities that does not bind any restriction over variables
C. Arithmetic equations that impose restrictions over variables
D. Arithmetic equations that discard constraints over the given variables

ugcnetcse-dec2015-paper3 optimization

Answer key

32.5.5 Optimization: UGC NET CSE | December 2015 | Part 3 | Question: 52

A basic feasible solution of a linear programming problem is said to be ______ if at least one of the basic variable is zero

A. generate B. degenerate C. infeasible D. unbounded

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Answer key

32.5.6 Optimization: UGC NET CSE | June 2012 | Part 3 | Question: 46

The feasible region represented by the constraints $x_1 - x_2 \leq 1, x_1 + x_2 \geq 3, x_1 \geq 0, x_2 \geq 0$ of the objective function Max $Z = 3x_1 + 2x_2$ is

A. A polygon
B. Unbounded feasible region
C. A point
D. None of these

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Answer key

32.5.7 Optimization: UGC NET CSE | June 2014 | Part 3 | Question: 58

Which of the following special cases does not require reformulation of the problem in order to obtain a solution ?

A. Alternate optimality
B. Infeasibility
C. Unboundedness
D. All of the above

ugcnetjune2014iii optimization

Answer key

32.5.8 Optimization: UGC NET CSE | Junet 2015 | Part 3 | Question: 69

Given the following statements with respect to linear programming problem:

S1: The dual of the dual linear programming problem is again the primal problem

S2: If either the primal or the dual problem has an unbounded objective function value, the other problem has no feasible solution

S3: If either the primal or the dual problem has a finite optimal solution, the other one also possess the same, and the optimal value of the objective functions of the two problems are equal.

Which of the following is true?

A. S1 and S2 B. S1 and S3 C. S2 and S3 D. S1, S2 and S3

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Answer key

32.6 Transportation Problem (5)

32.6.1 Transportation Problem: UGC NET CSE | December 2012 | Part 3 | Question: 28

The initial basic feasible solution to the following transportation problem using Vogel's approximation method is

	D_1	D_2	D_3	D_4	**Supply**
S_1	1	2	1	4	30
S_2	3	3	2	1	50
S_3	4	2	5	9	20
Demand	20	40	30	10	

A. $x_{11} = 20, x_{13} = 10, x_{21} = 20, x_{23} = 20, x_{24} = 10, x_{32} = 10$, Total cost $= 180$
B. $x_{11} = 20, x_{12} = 20, x_{13} = 10, x_{22} = 20, x_{23} = 20, x_{24} = 10$, Total cost $= 180$

C. $x_{11} = 20, x_{13} = 10, x_{22} = 20, x_{23} = 20, x_{24} = 10, x_{32} = 10$, Total cost $= 180$
D. None of the above

ugcnetcse-dec2012-paper3 optimization transportation-problem

Answer key

32.6.2 Transportation Problem: UGC NET CSE | December 2015 | Part 3 | Question: 53

Consider the following conditions:

i. The solution must be feasible, i.e. it must satisfy all the supply and demand constraints
ii. The number of positive allocations must be equal to $m+n-1$, where m is the number of rows and n is the number of columns
iii. All the positive allocations must be in independent positions

The initial solution of a transportation problem is said to be non-degenerate basic feasible solution if it satisfies:

A. i and ii only
B. i and iii only
C. ii and iii only
D. i, ii and iii

ugcnetcse-dec2015-paper3 optimization transportation-problem

Answer key

32.6.3 Transportation Problem: UGC NET CSE | December 2015 | Part 3 | Question: 54

Consider the following transportation problem:

Factories	Stores					
		I	II	III	IV	Supply
	A	4	6	8	13	50
	B	13	11	10	8	70
	C	14	4	10	13	30
	D	9	11	13	8	50
	Demand	25	35	105	20	

Factories	**Stores**					
		I	**II**	**III**	**IV**	**Supply**
	A	4	6	8	13	50
	B	13	11	10	8	70
	C	14	4	10	13	30
	D	9	11	13	8	50
	Demand	25	35	105	20	

The transportation cost in the initial basic feasible solution of the above transportation problem using Vogel's Approximation method is

A. 1450
B. 1465
C. 1480
D. 1520

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Answer key

32.6.4 Transportation Problem: UGC NET CSE | June 2014 | Part 3 | Question: 60

The initial basic feasible solution of the following transportion problem:

		Destination			
		D_1	D_2	D_3	Supply
Origins	O_1	2	7	4	5
	O_2	3	3	1	8
	O_3	5	4	7	7
	O_4	1	6	2	14
	Demand	7	9	18	

is given as

5		
		8
	7	
2	2	10

then the minimum cost is

A. 76 B. 78 C. 80 D. 82

ugcnetjune2014iii optimization transportation-problem

Answer key

32.6.5 Transportation Problem: UGC NET CSE | Junet 2015 | Part 3 | Question: 68

Consider the following transportation problem:

$\longrightarrow$ **Warehouse**

$\downarrow$ **Factory**

	W_1	W_2	W_3	Supply
F_1	16	20	12	200
F_2	14	8	18	160
F_3	26	24	16	90
Demand	180	120	150	

The initial basic feasible solution of the above transportation problem using Vogel's Approximation method (VAM) is given below:

$\longrightarrow$ **Warehouse**

$\downarrow$ **Factory**

	W_1	W_2	W_3	Supply
F_1	16(140)	20	12(60)	200
F_2	14(40)	8(120)	18	160
F_3	26	24	16(90)	90
Demand	180	120	150	

The solution of the above problem:

A. is degenerate solution
B. is optimum solution
C. needs to improve
D. is infeasible solution

ugcnetcse-june2015-paper3 transportation-problem optimization

Answer key

Answer Keys

32.1.1	B	32.1.2	A	32.2.1	A	32.3.1	D	32.4.1	C
32.5.1	C	32.5.2	D	32.5.3	B	32.5.4	A	32.5.5	A
32.5.6	B	32.5.7	A	32.5.8	D	32.6.1	D	32.6.2	A
32.6.3	B	32.6.4	A	32.6.5	B				